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**DEPARTMENT OF PHYSICS**

# Notes

**M.Sc Part II**

**SOLID STATE PHYSICS  
COMPULSORY**

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Phy. M.Sc. Part 2 E-1403 (0342-5969106)

Chapter # 06

(A)

M. ISHAQ  
E-1403

20/10/14

### Free Electron Fermi Gas :-

⇒ Energy level in one dimension.  
OR

⇒ An electron in an infinite deep potential well.

Consider a free electron gas in one dimension, taking account of Quantum theory and of the Pauli Principle.

We relate free electron gas theory with electron in an infinite deep potential well because on it ( $e^-$  in an infinite deep potential well) free electron approximation (i.e. ion core do not affect the electron) and independent electron approximation (i.e. electrons do not interact with each other) can be applied. So,

now consider an electron of mass ' $m$ ' is confined to a length ' $L$ ' by infinite barrier i.e.

$$V(x) = 0$$

$$0 < x < L$$

$$V(x) = \infty$$

$$x < 0, x > L$$

The wavefunction  $\psi_n(x)$  of electron is



(B)  
a solution of the Schrödinger equation.

$$H \psi_n(x) = E_n \psi_n(x)$$

where

$$H = E + V$$

$$H = \frac{1}{2} m v^2 + V$$

$$H = \frac{m^2 v^2}{2m} + V$$

$$H = \frac{p^2}{2m} + V$$

$V(x) = -V$

$V(x) = 0$

0

L

with the neglect of potential energy we will have

$$H = \frac{p^2}{2m}$$

So

$$\frac{p^2}{2m} \psi_n(x) = E_n \psi_n(x)$$

In Quantum theory 'p' may be represented as " $-i\hbar \frac{d}{dx}$ ", so

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi_n(x)}{dx^2} = E_n \psi_n(x) \quad (1)$$

where ' $E_n$ ' is the energy of an electron in the nth orbit. This is one-electron time independent S.W.E. Each solution

(E)

multiplying both sides by  $-\frac{\hbar^2}{2m}$

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi_n(x)}{dx^2} = -\left(\frac{n\pi}{L}\right)^2 \frac{\hbar^2}{2m} \psi_n = (E_n) \psi_n$$

Comparing equation (1) & (3), we get

$$E_n = \frac{\hbar^2}{2m} \left(\frac{n\pi}{L}\right)^2$$

$$E_n = \frac{\hbar^2}{2m} k_n^2 \quad (4)$$

$$E_1 = \frac{\hbar^2}{2m} \frac{\pi^2}{L^2}$$

$$E_2 = \frac{\hbar^2}{2m} \frac{4\pi^2}{L^2}$$

$$E_3 = 4E_1$$

$$E_4 = 9E_1$$

$$\vdots$$

$$\vdots$$

This is the energy of free electron.

We want to accommodate

'N' electrons on the line.

According to Pauli exclusion principle, no two electrons

can have all their quantum

numbers identical. That is each orbital can be occupied at most one electron. This applies to electrons in atoms, molecules or solids.

In a linear solid, the quantum numbers of a conduction electron orbital are 'n' and 'm<sub>s</sub>', where 'n' is any positive integer and magnetic quantum number 'm<sub>s</sub>' =  $\pm \frac{1}{2}$ , according



(F)  
to spin orientation. A pair of orbitals labeled by the quantum 'n' can accommodate two electrons, One with spin up and one with spin down.

## Free Electron Fermi Gas:-

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If there are six electrons, then in the ground state of the system, the filled orbitals are those given in the table.

| n | $m_s$ | Electron Occupancy |
|---|-------|--------------------|
| 1 | ↑     | 1                  |
| 1 | ↓     | 1                  |
| 2 | ↑     | 1                  |
| 2 | ↓     | 1                  |
| 3 | ↑     | 1                  |
| 3 | ↓     | 1                  |
| 4 | ↑     | 0                  |
| 4 | ↓     | 0                  |

It means that, if we <sup>have</sup> six electrons, then three energy levels will be filled, because more than two electrons can not exist in one energy level. Similarly if we have 100 electrons, then 50 energy levels will be filled.

(c)

of this equation is called an orbital. As we know that outside '0' and 'L' wavefunction  $\psi(x)$  is zero, so because of continuity of the wavefunction at boundaries, wavefunction should also vanish. So

$$\psi(0) = 0 \quad , \quad \psi(L) = 0$$

These are called boundary conditions. It is to be noted that solutions of S.W.E are stationary state solutions because stationary waves are produced in a well.

The boundary conditions are satisfied if wavefunction is sine like with an integral number 'n' of half wave lengths between '0' and 'L'.

So

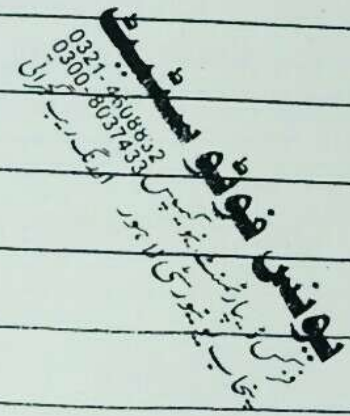
$$\psi_n(x) = A \sin K_n x$$

As

$$K_n = \frac{2\pi}{\lambda_n}$$

$$\psi_n(x) = A \sin\left(\frac{2\pi}{\lambda_n}\right) x$$

putting values of  $\lambda_n$

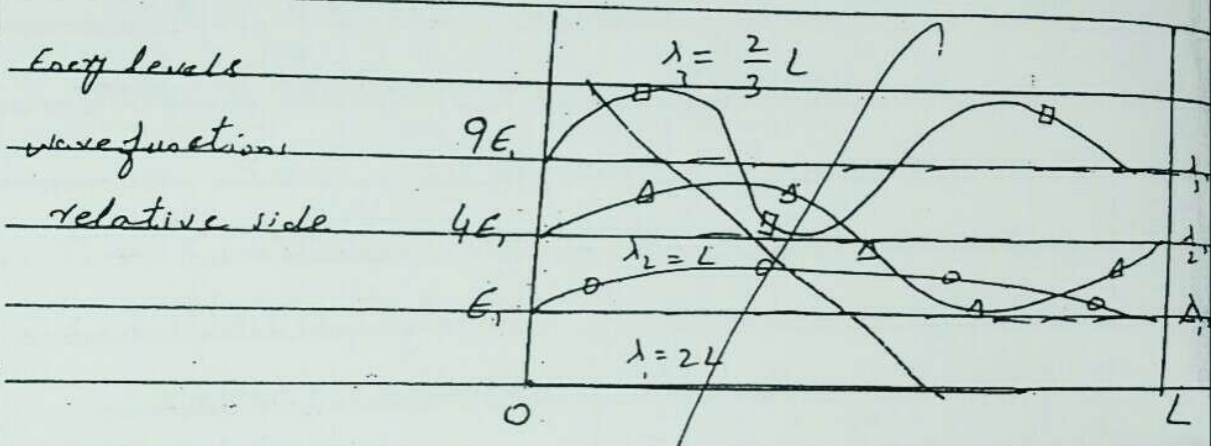




(d)

$$\psi_n(x) = A \sin\left(\frac{2n\pi x}{2L/n}\right)$$

$$\psi_n(x) = A \sin\left(\frac{n\pi x}{L}\right) \quad (2)$$



From figure,

$$L = 1 \cdot \frac{\lambda_1}{2}$$

$$L_n = \frac{n \lambda_n}{2}$$

$$L = 2 \cdot \frac{\lambda_2}{2}$$

$$\Rightarrow \lambda_n = \frac{2L}{n}$$

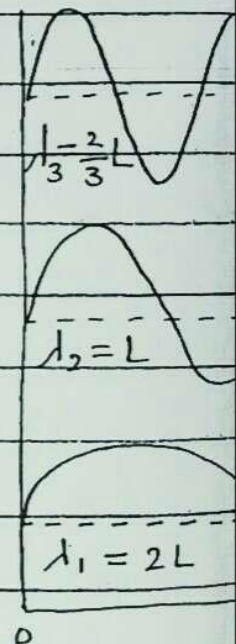
$$L = 3 \cdot \frac{\lambda_3}{2}$$

$$L_n = \frac{n \lambda_n}{2} \text{ From equation (1)}$$

$$\frac{d\psi_n(x)}{dx} = A \left(\frac{n\pi}{L}\right) \cdot \cos\left(\frac{n\pi x}{L}\right)$$

$$\frac{d^2\psi_n(x)}{dx^2} = -A \left(\frac{n\pi}{L}\right)^2 \sin\left(\frac{n\pi x}{L}\right)$$

$$\frac{d^2\psi_n(x)}{dx^2} = -\left(\frac{n\pi}{L}\right)^2 \psi_n(x)$$



(4)

The last energy level is called "fermi level" and is written as,

$$n_f = N/2$$

∴

The fermi energy ' $E_f$ ' is defined as the energy of the top most filled level in the ground state of 'N' electron system.

putting  $n = n_f$  in Equation (4) we have

$$E_f = \frac{\hbar^2}{2m} \left( \frac{n_f \pi}{L} \right)^2$$

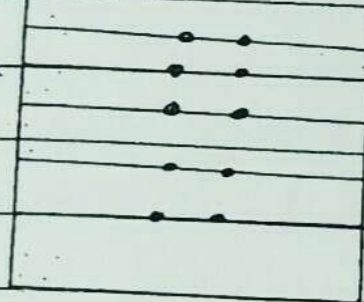
$$\because E_n = \frac{\hbar^2}{2m} k_n^2$$

$$E_n = \frac{\hbar^2}{2m} \left( \frac{n\pi}{L} \right)^2$$

$$E_f = \frac{\hbar^2}{2m} \left( \frac{N\pi}{2L} \right)^2 \quad \text{--- (5)}$$

Figure:-

$E_f$  ↑



Occupation of energy levels according to Pauli Exclusion principle.

### ✓ Effect of Temperature on Fermi Dirac Distribution Function:-

Electrons in solids obey Fermi Dirac statistics, in the development of this type of statistics, one can consider



the indistinguishability of electrons, their wave nature and the Pauli Exclusion principle.

The ground state is the state of ' $n$ ' electrons, system at the absolute zero. What happens as the temperature is increased. This is standard problem in the elementary statistical mechanics, and the solution is given by Fermi Dirac distribution function.

The kinetic energy of electrons increases as the temperature is increased. Some energy levels are occupied which were vacant at absolute zero and some energy levels were vacant which were occupied at absolute zero.

The FDDF gives the probability that an orbital at energy ' $E$ ' will be occupied in an ideal electron gas in thermal equilibrium.

$$f(E) = \frac{1}{e^{(E-E_F)/k_B T} + 1}$$

At absolute zero, the distribution function

has the following properties.

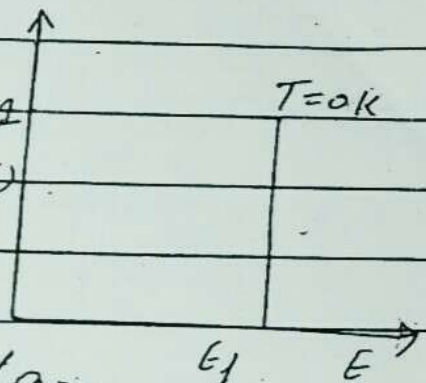
$$f(E) = 1 \quad \text{for all values of } E < E_f$$

$$f(E) = 0 \quad \text{for all value of } E > E_f$$

That is, all levels below  $E_f$  are completely filled and all those levels above  $E_f$  are completely vacant. This function is plotted in the figure below;

In general,  $f(E)$  has value 1 between '0' and unity  $f(E)$  i.e. at 0K, the DF

takes the simple rectangular form or step function.



When the system is heated ( $T > 0K$ ), thermal energy excites the electrons, but this energy is not shared equally by all the electrons as would be the case in the classical treatment, because the electrons lying well below the Fermi level ' $E_f$ ' can not absorb energy. If they did so, they would

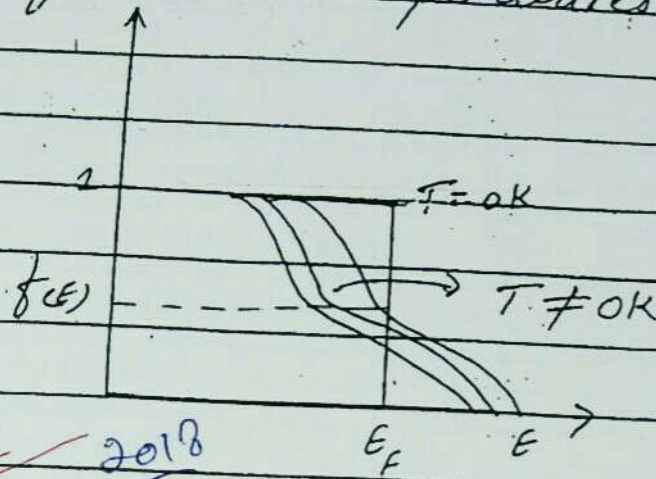


move to a higher level, which would be already occupied, and hence the pauli exclusion principle would be violated.

The energy which an electron may absorb thermally is of the order of  $k_B T$  ( $= 0.025 \text{ eV}$  at room Temp) which is much smaller than, this being of the order of  $5 \text{ eV}$ .

Therefore, only those electrons close to the fermi level can be excited, because the levels above  $E_f$  are empty and hence when those electrons move to a higher level, there is no violation of the exclusion principle. Thus only these electrons, which are the small fraction of the total number are capable of being thermally excited. For example, at  $T \neq 0 \text{ K}$ , there is some probability  $f(E)$  that states above  $E_f$  are filled and there is corresponding probability  $[1 - f(E)]$  that states below  $E_f$  are empty. The fermi level is symmetrical about

$E_F$  for all temperatures

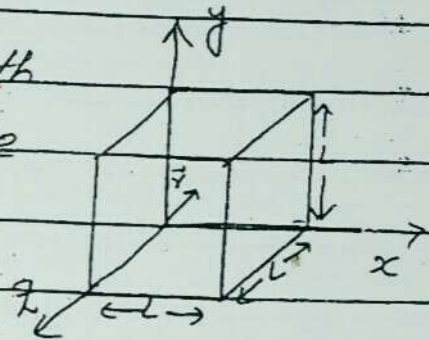


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## Free Electron Gas In Three Dimensions:-

Consider a cube and length of each side of cube be  $L$ .

For this cube, we have



$V(\vec{r}) = 0$  inside the box and  $V(\vec{r}) = \infty$  outside the cube. The free particle S.W.E in three-dimensional is,

$$-\frac{\hbar^2}{2m} \left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \psi_k(\vec{r}) = E_k \psi_k(\vec{r}) \quad \text{--- (1)}$$

If the electron is confined to a cube of edge 'L', the wavefunction is the standing waves.

$$\psi_k(\vec{r}) = A \sin\left(\frac{n_x \pi x}{L}\right) \sin\left(\frac{n_y \pi y}{L}\right) \sin\left(\frac{n_z \pi z}{L}\right) \quad \text{--- (2)}$$



where  $n_x, n_y, n_z$  are positive integers.  
This is the solution of stationary type waves in three dimension.

Now we want that electron should be confined in the cube and its waves should be traveling waves rather than stationary waves. To get this, we use periodic boundary conditions which are also called Bohr-Von Karman boundary conditions.

$$\text{i.e. } \psi(x+L, y, z) = \psi(x, y, z) \quad \text{--- (a)}$$

$$\psi(x, y+L, z) = \psi(x, y, z) \quad \text{--- (b)}$$

$$\psi(x, y, z+L) = \psi(x, y, z) \quad \text{--- (c)}$$

i.e. wavefunctions are periodic with period  $L$ .

Now the solution (2) is not valid because it is the solution of stationary waves.

Now we write the solution of traveling waves

$$\text{i.e. } \psi_{\vec{k}}(\vec{r}) = e^{i\vec{k} \cdot \vec{r}} \quad \text{--- (3)}$$

$$\psi_{\vec{k}}(\vec{r}) = e^{i(k_x x + k_y y + k_z z)} \quad \text{--- (4)}$$

# Solutions of traveling waves.

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$$k_x = 0, \pm \frac{2\pi}{L}, \pm \frac{4\pi}{L}, \dots = \frac{n_x 2\pi}{L}$$

$$n_x = 0, \pm 1, \pm 2, \dots$$

$$k_y = 0, \pm \frac{2\pi}{L}, \pm \frac{4\pi}{L}, \dots = \frac{n_y 2\pi}{L}$$

$$n_y = 0, \pm 1, \pm 2, \dots$$

$$k_z = 0, \pm \frac{2\pi}{L}, \pm \frac{4\pi}{L}, \dots = \frac{n_z 2\pi}{L}$$

$$n_z = 0, \pm 1, \pm 2, \dots$$

Now we want to confirm that the equation (4) with boundary conditions (a), (b), (c) satisfies the S.W.E. For simplicity, we consider only x-component

$$\text{i.e. } x = x + L$$

$$\psi_k(x+L) = e^{ik_x(x+L)}$$

$$\psi_k(x+L) = e^{i\left(\frac{n_x 2\pi}{L}\right)(x+L)}$$

$$\psi_k(x+L) = e^{\frac{i 2\pi n_x \cdot x}{L}} \cdot e^{i 2\pi x \cdot n_x}$$

$$\psi_k(x+L) = e^{\frac{i n_x 2\pi \cdot x}{L}} \left( \cos 2\pi n_x x + i \sin 2\pi n_x x \right)$$

$$\psi_k(x+L) = e^{\frac{i n_x 2\pi \cdot x}{L}} \left( \cos 2\pi n_x x + i \sin 2\pi n_x x \right)$$



$$\psi_K(x+L) = e^{iK_x \cdot x}$$

$$\psi_K(x+L) = \psi_K(x) \quad \text{Satisfied}$$

Now we want to find energy ' $E_K$ ' of the orbital with wavevector ' $K$ ' double differentiate  $\psi_K(\vec{r})$  and put it in S.W.E, So we have, (3)

$$\psi_K(\vec{r}) = e^{i\vec{K} \cdot \vec{r}}$$

$$\psi_K(\vec{r}) = e^{(iK_x x + iK_y y + iK_z z)} \quad \text{Diff w.r.t } x$$

$$\frac{\partial \psi_K(\vec{r})}{\partial x} = iK_x e^{i\vec{K} \cdot \vec{r}}$$

$$\frac{\partial^2 \psi_K(\vec{r})}{\partial x^2} = -K_x^2 e^{i\vec{K} \cdot \vec{r}} \quad \text{--- (I)}$$

Similarly

$$\frac{\partial^2 \psi_K(\vec{r})}{\partial y^2} = -K_y^2 e^{i\vec{K} \cdot \vec{r}} \quad \text{--- (II)}$$

$$\frac{\partial^2 \psi_K(\vec{r})}{\partial z^2} = -K_z^2 e^{i\vec{K} \cdot \vec{r}} \quad \text{--- (III)}$$

Adding (I) + (II) + (III)

$$\left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \psi_K(\vec{r}) = - (K_x^2 + K_y^2 + K_z^2) e^{i\vec{K} \cdot \vec{r}}$$

$$\left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \psi_K(\vec{r}) = -K^2 \psi_K(\vec{r})$$

put it in equation (2)

$$\frac{\hbar^2 k^2}{2m} (\hbar^2 k^2 \psi_k(\vec{r})) = E_k \psi_k(\vec{r})$$

$$E_k = \frac{\hbar^2 k^2}{2m}$$

that is the energy of electron where  $k = \sqrt{k_x^2 + k_y^2 + k_z^2}$  the magnitude of the wavevector ' $k$ ' is related to wavelength ' $\lambda$ ' by  $k = 2\pi/\lambda$ .

⇒ To find linear momentum, as we know that linear momentum ' $\vec{p}$ ' may be represented in Q.M by the operator

$$\vec{p} = -i\hbar \vec{\nabla}$$

So,

$$\vec{p} \psi_k(\vec{r}) = -i\hbar \vec{\nabla} \psi_k(\vec{r})$$

$$= -i\hbar \left( \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) e^{i\vec{k} \cdot \vec{r}}$$

$$\vec{p} \psi_k(\vec{r}) = -i\hbar (i\hbar e^{i\vec{k} \cdot \vec{r}})$$

$$\vec{p} = \hbar \vec{k}$$

So the plane wave solution  $\psi_k(\vec{r})$  is an eigen function of linear

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momentum operator  $\vec{p}$  with the eigenvalue  $\hbar k$ .

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### ✓ Fermi Sphere:-

In the ground state of a system of  $N$  free electrons, the occupied orbitals may be represented as points inside the sphere in  $k$ -space. This sphere is known as Fermi sphere, and its surface is known as Fermi surface. The energy at the surface is the Fermi energy i.e. maximum energy electrons lie on the surface of the sphere and Fermi energy is given by

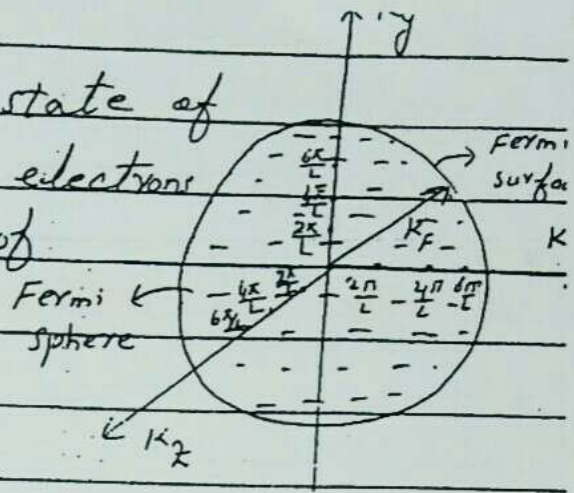
$$E_f = \frac{\hbar^2 k_f^2}{2m}$$

where ' $k_f$ ' (Fermi wave vector) is radius of Fermi sphere.

All the points inside the sphere are completely filled whereas all the points outside the sphere are empty because they correspond to energies greater than ' $E_f$ ' which are unoccupied at  $T=0K$ .

Figure:-

In the ground state of a system of  $N$  free electrons the occupied orbitals of the system fill a sphere of radius  $K_F$  where



$E_F = \frac{\hbar^2 K_F^2}{2m}$  is the energy of an

electron with wavevector  $K_F$ .

The particle velocity in orbital  $K$  is given by

$$V_F = \frac{\hbar K_F}{m} \quad (*) \quad E_F = \frac{\hbar^2 K_F^2}{2m}$$

& Fermi velocity is given by

$$\Rightarrow \hbar^2 K_F^2 = 2m E_F$$

$(*) \Rightarrow$

$$V_F = \frac{\hbar K_F}{m}$$

$$\hbar K_F = \sqrt{2m E_F}$$

$$= \frac{\sqrt{2m E_F}}{m}$$

$$V_F = \left( \frac{2 E_F}{m} \right)^{\frac{1}{2}}$$

The Fermi speed is very large if we

put  $E_F = 5 \text{ eV}$

$$\text{then } V_F = \left( \frac{2 \times 5 \times 1.6 \times 10^{-19}}{9 \times 10^{-31}} \right)^{\frac{1}{2}}$$

$$V_F = 10^6 \text{ m s}^{-1}, \text{ which is about one hundred}$$

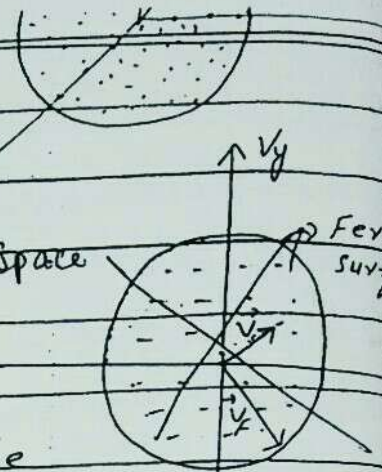


of the speed of light.

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## The Density of States:-

$v$ -space



In order to determine the actual number of electrons in a given energy state, it is necessary to know the number of states in the system which have energy under consideration multiplied by the probability distribution function. Therefore, if  $D(E)dE$  is the number of available quantum states in the unit energy range (density of states) and  $f(E)$  is the probability function of the electrons occupying a particular energy state ' $E$ ', then the actual number of electrons  $N(E)dE$  present in, so called free state in the unit range at any temperature is given by

$$N(E)dE = f(E) D(E) dE$$

We can also calculate, the total number of available quantum states, total number of electrons and density of

The total number of electrons in available Q. states is given by

$$N = \frac{V}{3\pi^2} \left( \frac{2mE}{\hbar^2} \right)^{3/2}$$

$$D(E) = \frac{dN}{dE} = \frac{V}{3\pi^2} \left( \frac{2m}{\hbar^2} \right)^{3/2} \cdot \frac{3}{2} (E)^{1/2}$$

$$D(E) = \frac{V}{2\pi^2} \left( \frac{2m}{\hbar^2} \right)^{3/2} E^{1/2}$$

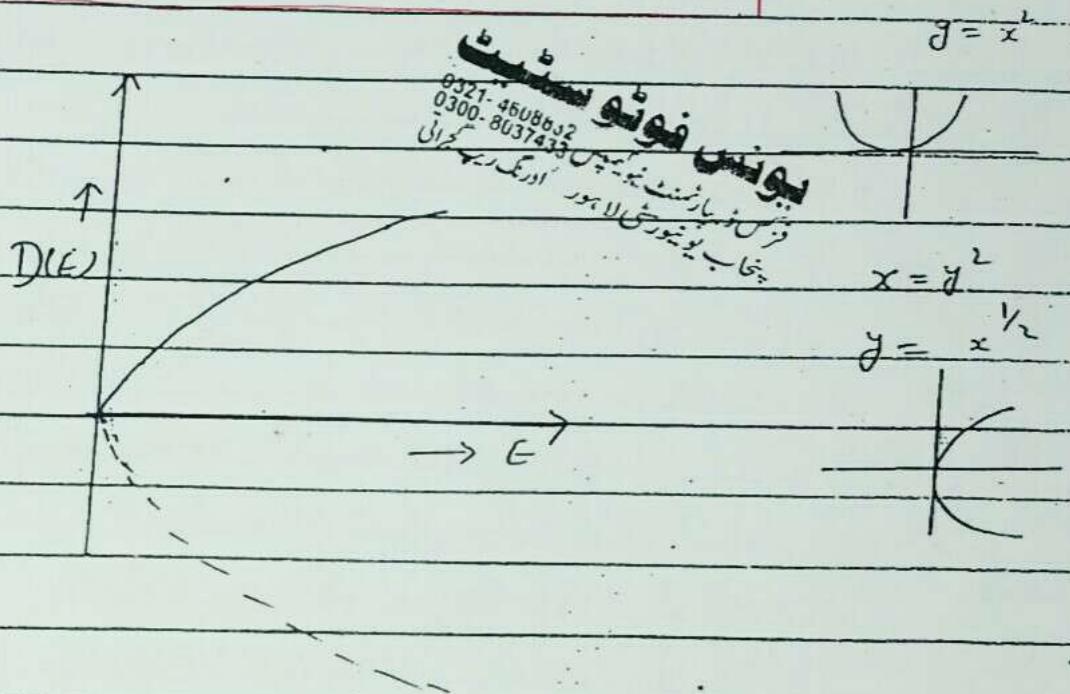
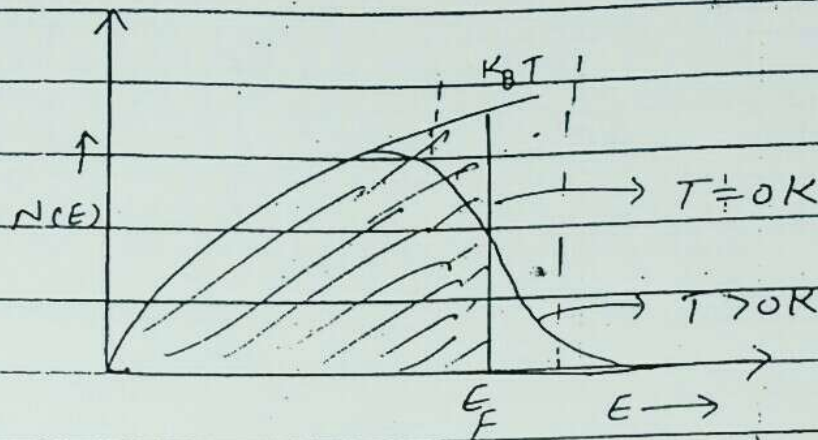


Figure:- Density of states as a function of energy for a free electron gas in three Dimension.

The graph shows that as energy increase the number of states increases, i.e. Density of states, tells us how many



many states of an energy are present. The Q. states increases quadratically with increase in energy.



$$N(E) = f(E) D(E)$$

Figure:-

The number of electrons or density of filled orbitals ( $N(E) = f(E) D(E)$ ) at 0K and at a finite temperature but  $K_B T$  is small in comparison with ' $E_F$ '. The shaded area represents the filled orbitals at absolute zero. The average energy is increased when the temperature is increased from '0' to  $T$ , for electrons are thermally excited from region 'i' to region 'j'. The expression for Density of states can be obtained more simply by taking natural log of ' $n$ '

i.e

$$\ln N = \frac{3}{2} \ln E + \text{Constant}$$

$$\frac{dN}{N} = \frac{3}{2} \frac{dE}{E}$$

$$D(E) = \frac{dN}{dE} = \frac{3}{2} \frac{N}{E}$$

i.e the number of orbitals per unit energy range at the fermi energy is the total number of conduction electrons divided by the fermi energy.

✓ Imp 2015 Heat Capacity of Electron Gas:- 11/11/14

The question that caused the greatest difficulty in the early development of electron theory of metals concerns the heat capacity of conduction electrons. Classical statistical mechanics predict that, a free particle should have a heat capacity of  $\frac{3}{2} K_B T$  where ' $K_B$ ' is Boltzmann Constant. If ' $N$ ' atoms, each give one valence electron to the electron gas and the electrons are freely mobile, then the electronic



Contribution to the heat capacity should be  $\frac{3}{2} k_B$ . But the observed electron contribution at room temperature is usually less than a1 of this value. This discrepancy distracted the early workers, such as Lorentz. How can the electrons participate in electrical conduction process as they were mobile, while not contributing to heat capacity?

The question was answered only upon the discovery of Pauli exclusion principle and the Fermi distribution function. Fermi found the correct equation and he wrote, "One recognizes that the specific heat capacity vanishes at absolute zero and - at low temperature it is proportional to absolute temperature when the specimen is heated from absolute zero not every electron gains an energy  $\sim k_B T$  as expected classically but only those electrons in orbitals within an energy range  $k_B T$  of the Fermi level are excited thermally. These electrons gain an energy which



is itself of the order of  $k_B T$ . This gives an immediate qualitative solution to the problem of the heat capacity of the conduction electron gas.

✓ So we conclude that only a fraction of  $\frac{k_B T}{E_F}$  of the electrons is affected

$$\text{i.e. } \frac{k_B T}{E_F} = \frac{k_B T}{k_B T_F} = \frac{T}{T_F}$$

If ' $N$ ' is total number of electrons only a fraction of the order of  $\frac{T}{T_F}$  can be excited thermally at temperature ' $T$ ', because only these lie within an energy range of the order of  $k_B T$  of the top of the energy distribution.

Each of these  $\frac{NT}{T_F}$  electrons has a thermal energy  $T_F$  of the order of  $k_B T$ . The total electronic thermal kinetic energy ' $U$ ' is of the order of

$$U = \left( \frac{NT}{T_F} \right) k_B T$$

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The electronic heat capacity is given by



$$C_{el} = \frac{\partial U}{\partial T}$$

$$C_{el} = N K_B \left( \frac{T}{T_F} \right)$$

and is directly proportional to ' $T$ ' in agreement with the experiments results. This is unlike the lattice heat capacity ' $C_L$ ', which is constant at high temperature and proportional to ' $T^3$ ' at low temperature.

At room temperature the ' $C_{el}$ ' is smaller than the classical value  $\frac{3}{2} N K_B$  by a factor of the order of 0.01 or less for  $T_F \sim 5 \times 10^4 \text{ K}$ , (a typical value correspond to  $E_F \sim 5 \text{ eV}$ )

Now we derive quantitative expression for the electronic heat capacity.

Consider a system of ' $N$ ' electron gas and when it is heated from ' $0$ ' to ' $T$ ', then change in energy is given as,

$$\Delta U = U(T) - U(0)$$

$$\text{Now } \int_0^\infty D(E) f(E) dE = N(E) \quad \begin{array}{l} \text{(Total number} \\ \text{of occupied state)} \\ \text{(i.e. electrons)} \end{array}$$

and  $\int_0^{\infty} E D(E) f(E) dE = \text{Total energy of the occupied states}$

At  $T=0K$  and states of energy  $E \leq E_F$  are occupied and so  $f(E) = 1$   
 So total energy at  $T=0K = \int_0^{E_F} E D(E) dE$

$$\Delta U = \int_0^{\infty} E D(E) f(E) dE - \int_0^{E_F} E D(E) dE \quad \text{--- (1)}$$

Since At temperature 'T' 17/11/14

$$N = \int_0^{\infty} D(E) f(E) dE \quad \text{--- (2)}$$

We can write it at  $T=0K$

$$N = \int_0^{E_F} D(E) dE \quad \text{--- (3)}$$

Multiplying eqn (2) & (3) by  $E_F$  and then Comparing, we have

$$\int_0^{\infty} E_F D(E) f(E) dE = \int_0^{E_F} E_F D(E) dE$$

integration by parts

$$\int_0^{E_F} E_F D(E) f(E) dE + \int_{E_F}^{\infty} E_F D(E) f(E) dE = \int_0^{E_F} E_F D(E) dE$$

$$\int_0^{E_F} E_F D(E) dE - \int_0^{E_F} E_F D(E) f(E) dE - \int_{E_F}^{\infty} E_F D(E) f(E) dE = 0$$



$$\int_0^{E_F} E_F D(E) (1 - f(E)) dE - \int_{E_F}^{\infty} E_F D(E) f(E) dE = 0 \quad (4)$$

Eqn (4) can be written as, adding  $\int_0^{E_F} E D(E) f(E) dE$ .

$$\Delta U = \underbrace{\int_0^{E_F} E D(E) f(E) dE}_{\text{Resolve}} - \int_0^{E_F} E D(E) dE + \int_0^{E_F} E_F D(E) (1 - f(E)) dE - \int_{E_F}^{\infty} E_F D(E) f(E) dE$$

$$\Delta U = \underbrace{\int_0^{E_F} (E_F D(E) (1 - f(E)) - E D(E)) dE}_{(1)} + \int_0^{E_F} E D(E) f(E) dE - \int_{E_F}^{\infty} E_F D(E) f(E) dE$$

or

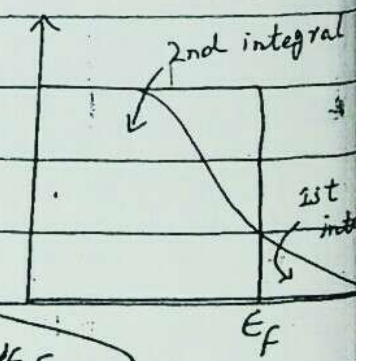
$$\Delta U = \int_{E_F}^{\infty} D(E) f(E) (E - E_F) dE + \int_0^{E_F} (E_F D(E) (1 - f(E)) - E D(E) (1 - f(E))) dE$$

or

$$\Delta U = \int_{E_F}^{\infty} D(E) f(E) (E - E_F) dE + \int_0^{E_F} D(E) (1 - f(E)) (E_F - E) dE \quad (5)$$

~~Statement~~

Now from eqn (5), we have



$$\Delta U = \int_{E_F}^{\infty} D(E) f(E) (E - E_F) dE + \int_0^{E_F} D(E) (E_F - E) dE + \int_0^{E_F} D(E) (E - E_F) f(E) dE$$

$$\Delta U = \int_0^{\infty} D(E) (E - E_F) f(E) dE \quad (6)$$

0.5%  $\int_{-\infty}^{\infty} B(E) dE$   میں نکالی گئی  
ہاں

Since

$$C_d = \frac{\partial (u)}{\partial T} \quad (\text{electronic heat capacity})$$

So

$$C_d = \int_0^{\infty} D(E) (E - E_F) \frac{df(E)}{dT} dE + 0 \quad (7)$$

The temperature dependent term is  $f(E)$ .

Since  $f(E)$  changes with temperature, near  $E_F$

So we can write,

$$D(E) \approx D(E_F)$$

$$C_d = D(E_F) \int_0^{\infty} (E - E_F) \frac{df}{dT} dE \quad (8)$$

Now

$$f(E) = \frac{1}{e^{\frac{E - E_F}{kT}} + 1}$$

Let

$$kT = \tau$$

$$f(E) = \left( e^{\frac{E - E_F}{\tau}} + 1 \right)^{-1}$$

$$\frac{df}{d\tau} = \frac{E - E_F}{\tau^2} \frac{e^{\frac{E - E_F}{\tau}}}{\left( e^{\frac{E - E_F}{\tau}} + 1 \right)^2} \quad (9)$$

Now let  $x = \frac{E - E_F}{\tau}$

So eqn (9) becomes

$$\frac{df}{d\tau} = \frac{x}{\tau} \frac{e^x}{(e^x + 1)^2} \quad (10)$$

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Since  $x = \frac{E - E_F}{T}$

$$xT = E - E_F$$

or  $E = xT + E_F$

$$dE = dxT$$

$$dE = kT dx$$

Also when  $E = 0$  ;  $x = -\frac{E_F}{T}$

when  $E = \infty$  ;  $x = \infty$

And  $T = kT \Rightarrow dT = k dT$

or  $dT = dT/k$

put all these values in eqn (8)

$$C_{el} = D(E_F) \int_{-\frac{E_F}{T}}^{\infty} (xT) \frac{d}{dT} (kT dx) k$$

$$= D(E_F) \int_{-\frac{E_F}{T}}^{\infty} xT \cdot \frac{x}{T} \cdot \frac{e^x}{(e^x + 1)^2} k^2 T dx$$

$$C_{el} = D(E_F) k^2 T \int_{-\frac{E_F}{T}}^{\infty} x^2 \frac{e^x}{(e^x + 1)^2} dx \quad (11)$$

So, eqn (11) becomes or  $E_F/kT$  will become large

$$C_{el} = k^2 T D(E_F) \int_{-\infty}^{\infty} \frac{x^2 e^x}{(e^x + 1)^2} dx \quad (12)$$

This is the standard integral,

$$\int_{-\infty}^{+\infty} \frac{x^2 e^x}{(e^x + 1)^2} dx = \frac{\pi^2}{3}$$

So eqn (12) becomes

$$C_d = \frac{\pi^2}{3} \cdot D(E_F) \cdot K^2 T \quad (3)$$

$$= \frac{\pi^2}{3} \cdot \frac{3N}{2E} \cdot K^2 T$$

$$C_d = \frac{\pi^2}{3} \left( \frac{3N}{2KT_F} \right) K^2 T = \frac{\pi^2}{2} NK \left( \frac{T}{T_F} \right) \quad (14)$$

The value of heat capacity calculated by Q-mechanics is better than that calculated by classical mechanics and its value is less as compared to classically calculated value.

Eqn (14) also can be written as,

$$C_d = \frac{\pi^2}{2} \frac{NK^2}{KT_F} \cdot T \quad \text{multiplying and dividing by 'K'}$$

$$C_d = \frac{\pi^2}{2} \frac{NK^2}{E_F} \cdot T$$

$$\text{But } E_F = \frac{\hbar^2 K_F^2}{2m}$$

$$\therefore = \frac{\pi^2}{2} NK^2 m T \quad (15) \nearrow$$

$$\frac{\pi^2}{2} \frac{NK^2 m T}{\hbar^2 K_F^2}$$



$$C_d = \gamma T \quad \text{where } \gamma = \frac{\pi^2 N k^2 m}{k^2 k_F^2}$$

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## Experimental Heat Capacity of Metals:-

The heat capacity of metals can be written as the sum of electron and phonon contribution. i.e.

$$C = \underbrace{\frac{3R\pi^4}{5} \left(\frac{T}{\theta_0}\right)^3}_{\text{Phonon Heat Capacity}} + \underbrace{\frac{\pi^2 N k^2 m T}{k^2 k_F^2}}_{\text{Electron heat capacity}} \quad \text{--- (I)}$$

$$C = \gamma T + AT^3$$

$$\frac{C}{T} = \gamma + TA^2 \quad \text{--- (II)}$$

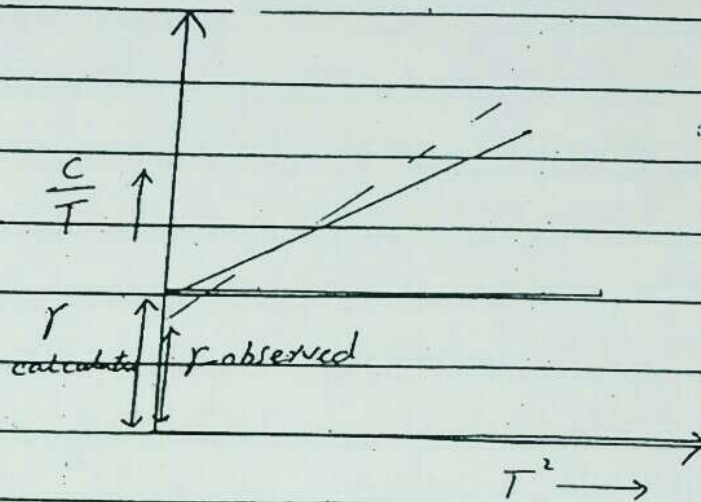
$$\text{where } \gamma = \frac{\pi^2 N k^2 m}{k^2 k_F^2}$$

$$\& A = \frac{3R\pi^4}{5\theta_0^3}$$

where  $A$  &  $\gamma$  are constants characteristic of material. As seen from (II); electron heat capacity is linear in ' $T$ ' and

is dominant at sufficiently low temperatures.

It is convenient to exhibit the experimental values of 'c' as a function of  $\frac{c}{T}$  versus " $T^2$ ".



From the figure, it is clear that observed value of 'r' is different from calculated value of 'r'.

As we know that 'r' depends on mass of electron as in eqn (I), so experimental mass and mass used to calculate 'r' is different.

So it is common practice to express the ratio of the observed to free electron values of electronic heat capacity as a ratio of thermal effective mass (mass of electron gas) " $m_{th}$ " to mass of



'e' m. So

$$\gamma_{(\text{observed})} = \frac{m_{th}}{m}$$

$\gamma_{(\text{free})}$

why  $\gamma$  differs:-

There are three effects for difference  $\gamma$

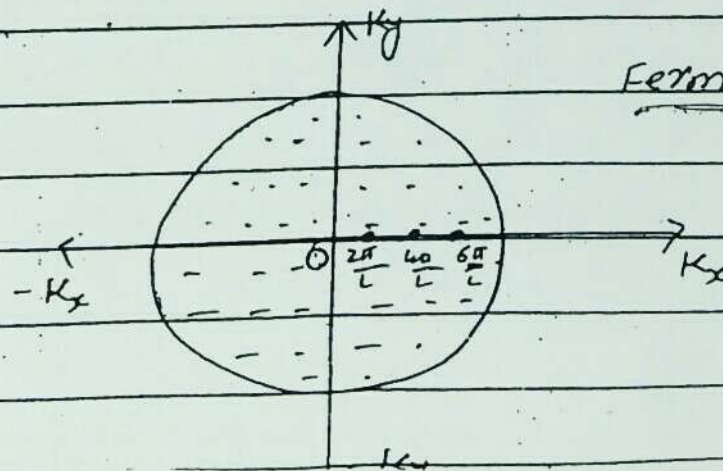
- (1) We have said that electrons are free (in free electron approximation), but actually they are not free, they move in periodic potential because they have interactions with other atoms also. The effective mass of electron in this potential is called the "band effective mass".
- (2) Electron tries to drag the ion-core with itself due to attraction between them. So effective mass of electron increases.
- (3) The interaction of conduction electrons with themselves ( $e^-e^-$  interaction). An electron causes an inertial reaction in the surrounding  $e^-$ -gas (the inertial reaction of  $e^-$ -gas), thereby increasing the effective mass of  $e^-$ .

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## Electrical Conductivity and Ohm's Law:-

There are some properties of metals which remain same using classical and Fermi models and electrical conductivity is one of them. Here we will see that how are the results modified when 'Fermi surface' is taken into account.

In the absence of electric field, the Fermi sphere is centered at the origin. The various electrons are all moving, some at very high speeds and they carry individual currents. But the total current of the system is zero, because for every orbital ' $k$ ', there is an occupied orbital at ' $-k$ ' and the sum of their currents is zero. Thus the total current vanishes due to pairwise cancellation of the electron currents.



Fermi sphere at  $t=0$



Figure:-

The Fermi sphere encloses the occupied electron orbital in  $\vec{k}$ -space in the ground state of the electron gas.

Now we will see, how momentum or wave vector of an electron is changed under the influence of electric field. The momentum of free electron is related to the wavevector by

$$m\vec{v} = \hbar \vec{k} \quad \text{--- (i)}$$

$$\text{As } \vec{F} = \frac{d\vec{p}}{dt}$$

$$\frac{d\vec{p}}{dt} = \hbar \frac{d\vec{k}}{dt}$$

$$F = \hbar \frac{d\vec{k}}{dt} \quad \text{--- (I)}$$

The force on electron in the presence of an electric field is given by

$$\vec{F} = -e\vec{E} \quad \text{--- (II)}$$

Comparing (I) & (II)

$$-e\vec{E} = \hbar \frac{d\vec{k}}{dt}$$

$$\frac{d\vec{k}}{dt} = \frac{-e\vec{E}}{\hbar}$$

Input

This is the rate of change of wavevector w.r.t 'time' in an electric field,  $\vec{E}$ .

To calculate change in ' $\vec{k}$ ' in time ' $t$ ' i.e. from  $t=0$  to  $t=t$

$$\int_0^t \frac{d\vec{k}}{dt} = -\frac{e\vec{E}}{\hbar} \int_0^t dt$$

$$\vec{k}(t) - \vec{k}(0) = -\frac{e\vec{E}t}{\hbar}$$

$$\delta \vec{k} = -\frac{e\vec{E}t}{\hbar}$$

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In the absence of collision, Fermi sphere moves in  $k$ -space at uniform rate by constant applied field  $\vec{E}$ .

Now

$$\delta \vec{k} = -\frac{e\vec{E}t}{\hbar} \quad \text{change in } \vec{k} \text{ in time } 't' \text{ in electric field } \vec{E}.$$

$$\hbar \delta \vec{k} = -e\vec{E}t \quad \text{change in momentum in time } 't' \text{ in electric field } \vec{E}.$$

$$\frac{\hbar \delta \vec{k}}{t} = -e\vec{E} \quad \text{rate of change of momentum in electric field } \vec{E}.$$

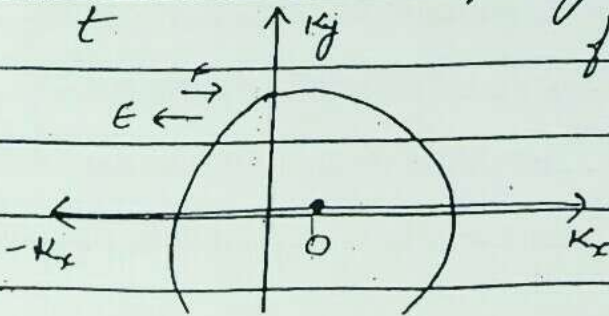




Figure:- Displacement of fermi sphere to an electric field at time  $t$ .  
sphere has origin at '0'.  
the whole fermi sphere will shift  
by  $\delta k = \frac{4\pi}{L}$

It means after some time, the fermi sphere will displace to a new centre in electric field  $\vec{E}$ .

When we apply electric field, then  $m\vec{v}_f - m\vec{v}_i$  is change in momentum for electron in metal and in fermi sphere, change in momentum is  $\hbar\delta\vec{k}$ , so

$$m\vec{v}_f - m\vec{v}_i = \hbar\delta\vec{k}$$

$$\vec{v}_f - \vec{v}_i = \frac{\hbar\delta\vec{k}}{m} = \frac{-e\vec{E}t}{m}$$

$$\vec{v}_f - \vec{v}_i = \vec{v}_d \rightarrow \text{drift velocity}$$

Now current starts in metals and momentum is not zero and fermi sphere shifts and now its momentum is also not zero. There exists some momentum because '3' & '4' momenta cancel each

Figure:- Displacement of fermi sphere to an electric field at time  $t$ .  
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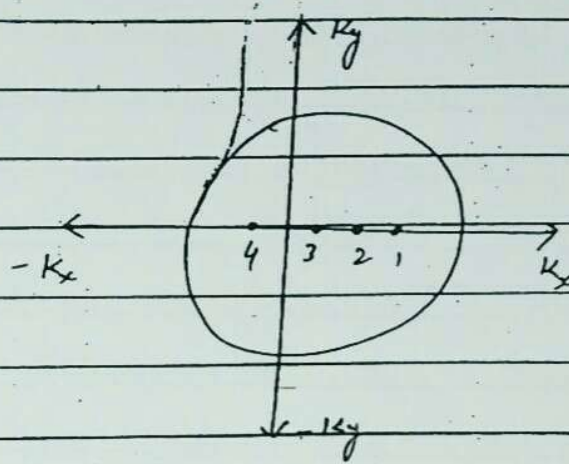
$$\vec{v}_f - \vec{v}_i = \frac{\hbar\delta\vec{k}}{m} = \frac{-e\vec{E}t}{m}$$

$$\vec{v}_f - \vec{v}_i = \vec{v}_d \rightarrow \text{drift velocity}$$

Now current starts in metals and momentum is not zero and fermi sphere shifts and now its momentum is also not zero. There exists some momentum because '3' & '4' momenta cancel each other.



other but to cancel '1' & '2' there is no momenta in opposite direction.



Now with the applied force,  $v_d$  25/11/14 increases and Fermi sphere shifts. It seems that both ' $v_d$ ' and ' $\delta K$ ' should go to infinity because every time with the increase in force, they increase but actually due to resistance in metals, ' $v_d$ ' does not go to infinity. In mean free time, it increases but when it collides with some ion, ' $v_d$ ' goes to zero i.e. between two collisions ' $v_d$ ' increases and at the time of collision, it goes to zero and due to this ' $v_d$ ' does not go to infinity but we get its average value and due to ' $v_d$ ', ' $\delta K$ ' limits itself. It means at every collision,  $v_d \rightarrow 0$  &

The actual picture of electrical conduction is thus quite different from the classical one, in which we assume that the ~~current~~ is carried equally by all electrons, each moving with a very small velocity ' $V_d$ '. The current is, in fact, carried by very few electrons only, all moving at high velocity. Both approaches lead to the same result, but the later is more accurate.

This can be seen from the fact that only the collision time for electrons at the Fermi ~~surface~~, ' $\tau_F$ ' appears in the expression for ' $\rho$ '.

### Experimental Electrical Resistivity of Metals:-

We know that resistivity of metal is defined as,

$$\rho = \frac{m}{Ne^2 \tau_F}$$

The 'resistivity ' $\rho$ ' of metal arises because electron undergoes a collision, only because the lattice is not perfect regular. The deviation from a perfect lattice can be grouped into two classes.



$\delta K = 0$ , So sphere stops with increase in ' $v_d$ ', ' $\delta K$ ' increases and sphere starts shifting/moving.

So sphere stops, ... then moves, then stops and so on, so

$$v_d = \frac{-eE\tau_F}{m} \quad \therefore t = \tau_F$$

because of

if in a constant electric field  $E$ , there are ' $n$ ' electrons of charge  $q = -e$  per unit ' $v_d$ ' the electric current density is,

$$J = -nev_d$$

putting value of ' $v_d$ ', we have

$$J = + \frac{ne^2 E \tau_F}{m}$$

where ' $\tau_F$ ' is the collision time of an electron at the Fermi sphere. The resulting conductivity is therefore given by

$$\sigma = \frac{ne^2 \tau_F}{m} \quad \therefore J = \sigma E$$

This is precisely the same as the result obtained classically except ' $\tau$ ' is replaced by ' $\tau_F$ '.

phonons. Particle nature of elastic waves are represented by phonons.

- (a) Lattice vibrations (phonons) of the atoms around their equilibrium positions due to thermal excitation of the ions.
- (b) All imperfections, such as foreign impurities or crystal defects.
- Therefore, the net resistivity is,

$$\rho = \rho_L + \rho_i$$

where

$\rho_L \rightarrow$  Resistivity caused by thermal phonons or lattice vibration.

$\rho_i \rightarrow$  Resistivity caused by scattering of electrons ( $e^-$ ) waves by defects that disturb the periodicity of the lattice.

The electrical resistivity of most metals is dominant at room temperature (or  $27^\circ\text{C}$ ) by collisions of the conducting electrons with lattice phonons or at liquid helium temperature ( $4\text{K}$ ) or absolute zero by collisions with impurity atoms and mechanical imperfections in the lattice as shown in the figure.



Define A group of geometric points, in which each point specifies the position of atom (or atoms), is called lattice.

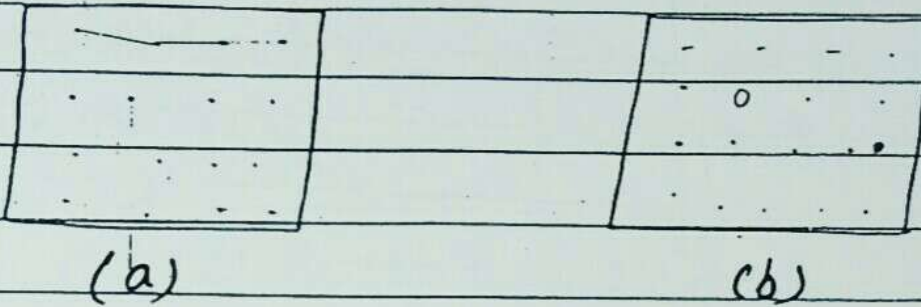


Figure:- electrical resistivity in most metals arises from collisions of electrons with irregularities in the lattice as in (a), by phonons in (b) by impurities or vacant lattice sites.

Rules from experiment:-

- (1) Often ' $\rho_i$ ' is independent of temperature and may vary from specimen to specimen.
- (2) ' $\rho_i$ ' is independent of defects but depends on temperature.

The above two rules are called

✓ Mattheissen's rule.

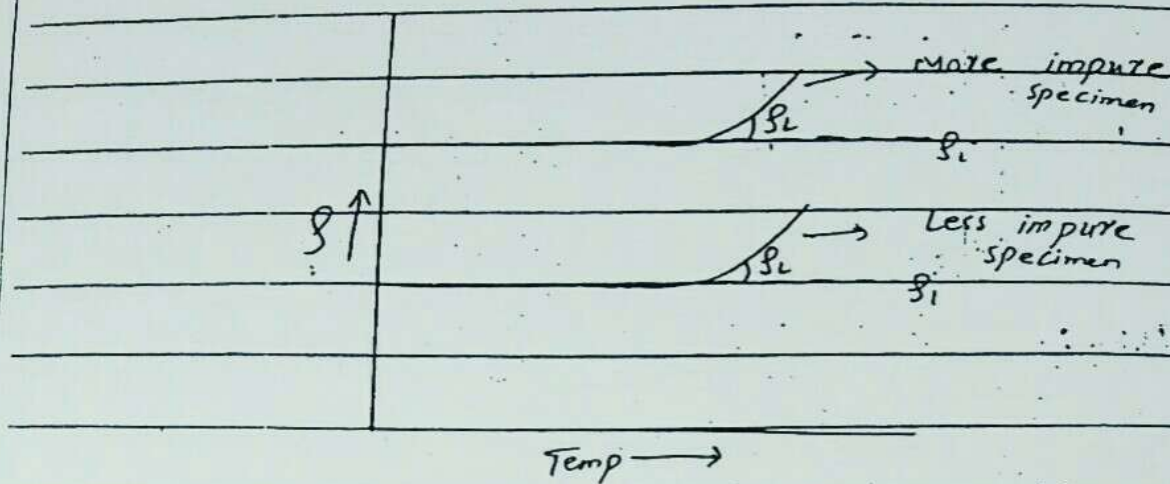
At  $T=0K$ , the whole resistivity is due to impurity because ' $\rho_i$ ' vanishes at  $T=0K$  i.e.

$$\rho = \rho_i(0) + 0$$

$\Rightarrow \rho_i(0)$  is called residual resistivity.

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The following figure shows the graph b/w ' $\rho$ ' & ' $T$ ' for any metal with two samples with respect to temperature.

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From the graph, it is clear that ' $\rho_i$ ' remains constant with the change in temperature whereas ' $\rho_L$ ' changes with temperature.

Also ' $\rho_L$ ' is small for impure specimen ' $\rho_i$ ' (It is as low as 1).

' $\rho_L$ ' is large for pure specimen.

' $\rho_i$ ' (This may be as large as  $10^6$ ).

Resistivity Ratio:-

The resistivity ratio of a specimen is usually defined as



i.e.

$$\text{Resistivity Ratio} = \frac{\text{Resistivity at room temperature}}{\text{Residual resistivity}}$$

$$= \frac{\rho_i(300\text{K}) + \rho_L(300\text{K})}{\rho_i(0\text{K})}$$

$$= \frac{\rho_i(0\text{K}) + \rho_L(300\text{K})}{\rho_i(0\text{K})}$$

$$\therefore \left\{ \begin{array}{l} \rho_L(300\text{K}) = \rho_L(0\text{K}) \\ \text{independent of} \\ \text{temperature} \end{array} \right.$$

$$\text{Resistivity ratio} = 1 + \frac{\rho_L(300\text{K})}{\rho_i(0\text{K})}$$

### Motion In A Magnetic Field:-

Application of magnetic field to a sample gives rise to several interesting effects arising from the conduction electrons.

The cyclotron resonance and the Hall effect are two which can be used to investigate the properties of conduction electrons.

### Cyclotron Resonance:-

Consider now the motion of electrons in a uniform magnetic field and electric field.

So Lorentz force on an electron is

$$\vec{F} = -e \left( \vec{E} + \frac{1}{c} \vec{v} \times \vec{B} \right)$$

There is also a friction force due to collision of electrons with rest of the medium. Let us assume that this friction force has the form  $-\frac{m^* \vec{v}}{\tau}$ , where ' $\vec{v}$ ' is the velocity of the electrons and ' $\tau$ ' is the collision time. Using Newton's law, we have

$$m^* \frac{d\vec{v}}{dt} = -e \left( \vec{E} + \frac{1}{c} \vec{v} \times \vec{B} \right) - \frac{m^* \vec{v}}{\tau}$$

where ' $m^*$ ' is the effective mass of electron. The effect of the collision as usual in friction or viscous forces tends to reduce the velocity to zero.

Therefore in steady state solution, where  $\frac{d\vec{v}}{dt} = 0$ , therefore above equation becomes

$$\frac{m^* \vec{v}}{\tau} = -e \left( \vec{E} + \frac{1}{c} \vec{v} \times \vec{B} \right)$$

where ' $\vec{v}$ ' is drift velocity of electron.



Now we have to find out the components of this velocity. For this, we consider a particular situation, in which  $\vec{B}$  is along  $z$ -axis and  $\vec{E}$  can be in any direction.

$$\frac{m^*}{\tau} (v_x \hat{i} + v_y \hat{j} + v_z \hat{k}) = -e \left[ E_x \hat{i} + E_y \hat{j} + E_z \hat{k} + \frac{1}{c} \{ (v_x \hat{i} + v_y \hat{j} + v_z \hat{k}) \times (B \hat{k}) \} \right]$$

$$v_x = \frac{-e}{m^*} \tau E_x - \frac{eB}{m^* c} v_y$$

$$v_y = \frac{-e}{m^*} \tau E_y + \frac{eB}{m^* c} v_x$$

$$v_z = \frac{-e}{m^*} \tau E_z$$

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These are the equations for the components of drift velocity when electric and magnetic field is applied and these equations hold only for steady state conditions.

where  $\omega_c = \frac{eB}{m^* c}$  is known as cyclotron

frequency and is defined as.

★ A magnetic field applied across a metallic slab causes electrons to move in a counter clockwise circular fashion in a plane normal to the field.

The frequency of this cyclotron motion is known as cyclotron frequency.  
So

$$V_x = \frac{-e\tau E_x}{m^*} - \omega_c \tau V_y$$

$$V_y = \frac{-e\tau E_y}{m^*} + \omega_c \tau V_x$$

$$V_z = -\frac{e\tau}{m^*} E_z$$

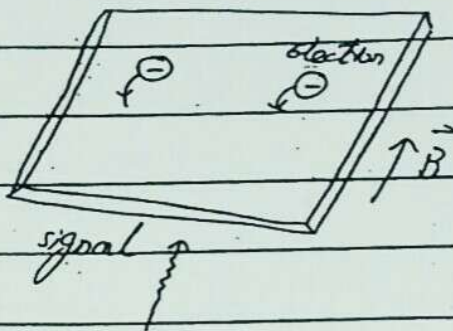


Figure:- Cyclotron motion.

Suppose now that electromagnetic signal is passed through the slab in a direction parallel to ' $\vec{B}$ ' as shown in the figure. The electric field of the signal acts on the electrons, and some of the energy in the signal is absorbed. The rate of absorption is greatest when the frequency of the signal is exactly equal to the frequency of the cyclotron.

$$\omega = \omega_c$$



40  
Cyclotron resonance is commonly used to measure the electron mass in metals and semiconductors. The cyclotron frequency is determined from the absorption curve and this value is substituted in eqn (A) to evaluate the effective mass. The accuracy with which  $m^*$  is determined depends upon the accuracy of ' $\omega_c$ ' and ' $B$ '.

One can measure the cyclotron frequency very accurately, particularly if one uses a laser beam, and therefore the accuracy of measurement of  $m^*$  is limited only by the accuracy of measurement of the magnetic field and its homogeneity across the sample.

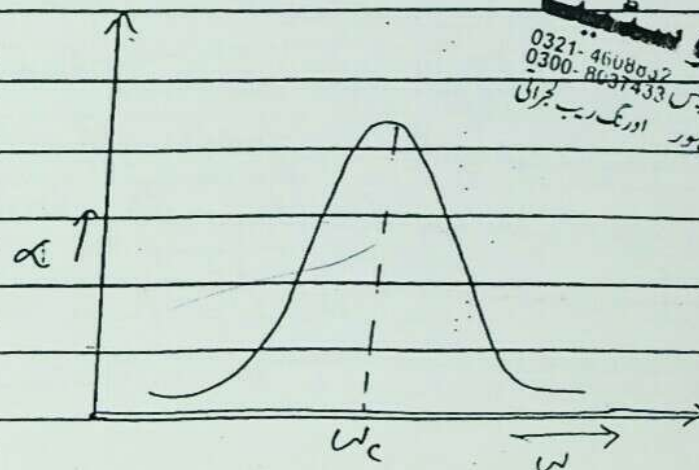
### ✓ The Hall effect:-

In 1897, Hall discovered that when a current is passed through a slab of material in the presence of transverse magnetic field, a small potential difference is established between the two opposite faces of the specimen which is perpendicular to both the direction of current flow and the direc

This is so, because when the condition holds true, each electron moves synchronously with the wave throughout the cycle and therefore, the absorption continues all through the cycle.

Thus the above equation is the condition for cyclotron resonance.

On the other hand, when the above equation is not satisfied, the electron is in phase with the wave through only a part of cycle, during which time it absorbs energy from the wave. In the remainder of the cycle, the electron is out of phase, and returns energy to the wave. The shape of the absorption curve as a function of frequency is shown in figure below.

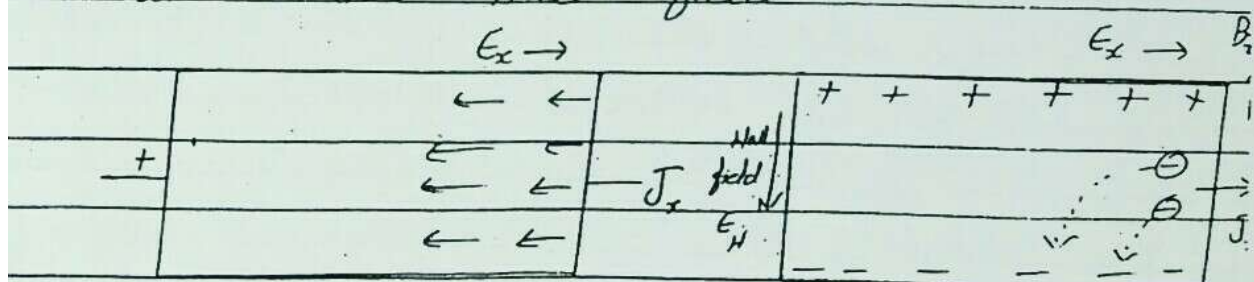


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of magnetic field. This effect is known as Hall effect and voltage so developed is "Hall voltage".

To see how this comes about, suppose an electric current flows in the  $x$ -direction and the magnetic field applied across the  $z$ -direction. Due to the magnetic field, the magnetic force  $\vec{F}_m = -e(\vec{v} \times \vec{B})$  causes the electron to bend downward as shown in the figure. As a result, electrons accumulate on the lower surface, producing a net negative charge there. Simultaneously, a net positive charge appears on the upper surface, because of the deficiency of electrons there. This combination of positive and negative surface charges creates a downward electric field, which is called the "Hall field".



- (a) An electric field is applied that causes current to flow in  $x$ -direction.
- (b) The deflection in the  $y$ -direction is caused by the magnetic field.



## Hall Effect:-

Let us evaluate this "Hall field". The Lorentz force " $F_L$ " which produces the charge accumulation in the first place is in the negative y-direction and has the value,

$$\vec{F}_L = -e v_x \vec{B} \quad \text{--- (*)}$$

Now the field created by the surface charges produces a force which opposes this Lorentz force.

The accumulation process continues until the Hall force completely cancels the Lorentz force.

Thus, in steady state

$$F_H = F_L$$

$$-e E_H = -e v_x B$$

or

$$E_H = v_x B$$

which is Hall field. It is convenient to express this in terms of measurable quantities and for this purpose, the velocity  $v_x$  is expressed in terms of current density

$$\vec{J}_x = -N e v_x$$

This leads to

$$E_H = -\frac{1}{ne} \vec{J}_x B$$



$$\vec{F} = -e(\vec{E} + \vec{v} \times \vec{B}) \quad \text{if } \vec{E} = -\vec{v} \times \vec{B}$$

$$\vec{F} = -e\vec{v} \times \vec{B} \Rightarrow \text{For } -ve \text{ } \vec{F} = -e\vec{v} \times \vec{B}$$

The Hall field is thus proportional to both the current and the magnetic field.

This proportionality constant that  $\frac{E_H}{J_x B}$  is known as Hall Constant, and is usually denoted by " $R_H$ ", therefore " $R_H$ " is written as,

$$R_H = -\frac{1}{ne}$$

Since

" $R_H$ " is inversely proportional to the electron concentration " $n$ ", it follows that we can determine " $n$ " by measuring the Hall field.

In fact, this is standard technique for determining the electron concentration. The technique is particularly valuable because apart from " $n$ " the only other quantity on which " $R_H$ " depends is the charge on the electron " $-e$ ".

which is fundamental physical constant whose value is known by accurately.

Another useful feature of the Hall constant is that its sign depends

on

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on the sign of current carriers.

Thus electrons being negatively charged lead to negative Hall constant.

If the conduction is by holes, then Hall constant will be positive. The sign of ' $R_H$ ' indicates the sign of carriers involved, which is very valuable information, particularly in the case of semiconductors.

### Thermal Conductivity of Metals:-

When the ends of metallic wire are at different temperatures, then heat flows from hot end to the cold end.

The basic experimental fact is that heat current ' $Q$ ' that is amount of thermal energy crossing a unit area per unit time, is proportional to the temperature gradient.

$$Q = K \frac{dT}{dx}$$

where ' $K$ ' is thermal conductivity.

In insulators, heat is carried entirely by phonons, but in metals heat



may be transported by both electrons and phonons.

The conductivity ' $K$ ' is therefore equal to the sum of two contributions

$$K = K_e + K_{ph}$$

where ' $K_e$ ' and ' $K_{ph}$ ' refer to electrons and phonons respectively.

In pure metals, the electronic contribution is dominant at all temperatures because of great concentration of electrons.

typically

$$K_{ph} \sim 10^{-2} K_e$$

But

in impure metals or in disordered alloys, the electron mean path is reduced by collision with impurity and phonon contribution may be comparable with the electronic contributions. Here we will discuss only thermal conductivity of pure metals. So, conductivity of phonons can be ignored.

The physical process by which heat conduction takes place via electrons is illustrated in the figure, below

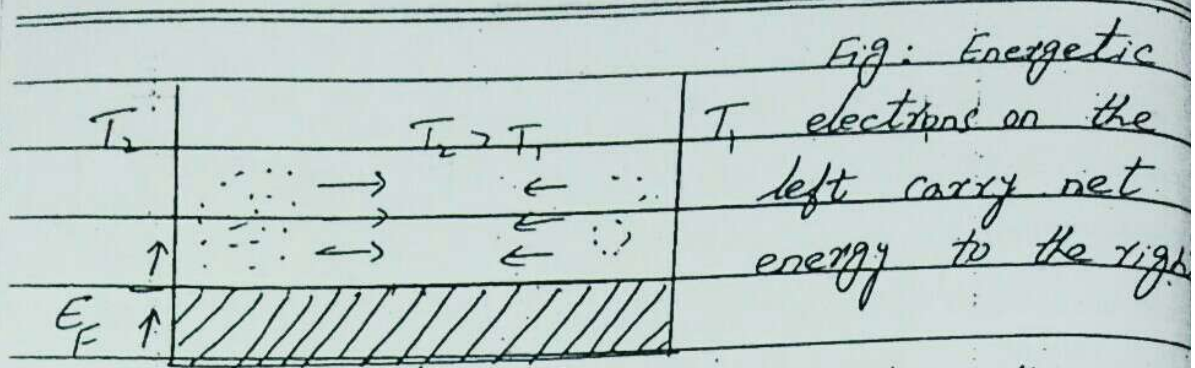


Figure:- The physical process for Thermal Conductivity.

Electrons at the hot end (to the left) travel in all directions, but a certain fraction travel to the right and carry energy to the (right) cold end. Similarly, a certain fraction of  $e^-$ s at the cold end (on the right) travel to the left, and carry energy to the hot end.

But electrons at the hot end are more energetic on the average than those on the right, a net energy is transported to the right, resulting in the current of heat.

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Note that heat is transported almost entirely by those electrons near the Fermi levels, because those well below the level cancel each other's



Contributions. It means that electrons at the Fermi surface play a primary role in transport phenomenon.

To evaluate the thermal conductivity 'K' quantitatively, we use the formula

$$K = \frac{1}{3} C_v v l$$

that is related to heat transport in insulators where 'C<sub>v</sub>' is specific heat per unit volume, 'v' is speed and 'l' is mean free path of the particles involved in the present case, where electrons are involved, we will consider electronic specific heat. In addition 'v' and 'l' should be replaced by 'v<sub>F</sub>' and 'l<sub>F</sub>' since only electrons at Fermi levels are effective.

Thus

$$K_{el} = \frac{1}{3} \left[ \frac{\pi^2 N K_B^2 T}{2 E_F} \right] v_F l_F$$

Noting that

$$E_F = \frac{1}{2} m v_F^2$$

&

$$l_F = v_F \tau_F$$

So 
$$K_{el} = \frac{\pi^2 n K_B^2 T \tau_F}{3m}$$

If we divide it by 'V', then we will get specific heat per unit volume

$$K_{el} = \frac{\pi^2 n K_B^2 T \tau_F}{V 3m}$$

$$K_{el} = \frac{\pi^2 n K_B^2 T \tau_F}{3m}$$

which expresses thermal conductivity in terms of electronic properties of metals.

As we know that

$$\delta = \frac{ne^2 \tau_F}{m}$$

$$\frac{K_{el}}{\delta} = \frac{\pi^2}{3} \left( \frac{K_B}{e} \right)^2 T$$

Since  $\frac{\pi^2}{3} \left( \frac{K_B}{e} \right)^2$  is a constant and is called Lorentz number. Its value should be same for all metals, because it depends only on the universal



constants ' $k_B$ ' and ' $e$ '

So,

$$L = \frac{\pi^2}{3} \left( \frac{k_B}{e} \right)^2 = 2.45 \times 10^{-8} \text{ watt-ohm/deg}^2$$

From above equation

$$\left[ \frac{K}{\delta} \propto T \right]$$

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This is called "Wiedmann Franz Law" which states that

"The ratio of thermal conductivity to electrical conductivity is directly proportional to the temperature with the value of constant of proportionality independent of particular metal".

problems:- 2:-

For sodium metal

$$\frac{N}{V} = 2.6 \times 10^{22} \text{ electrons/cm}^3$$

(a) Find  $E_F$

(b) Approximate value of electronic specific heat of Na, at room temperature.

Sol:- AS we know that

$$E_F = \frac{\hbar^2}{2m} \left( \frac{3\pi^2 N}{V} \right)^{2/3}$$

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